

DiffNorm-Contact HMR: Annotated Related Work & Literature Taxonomy

A Comprehensive Review of Parametric Models, Human Mesh Recovery, 3D Gaussian Splatting, Foundation Geometric Priors, and Physical Contact Dynamics

Author: Antigravity Research Intelligence Team

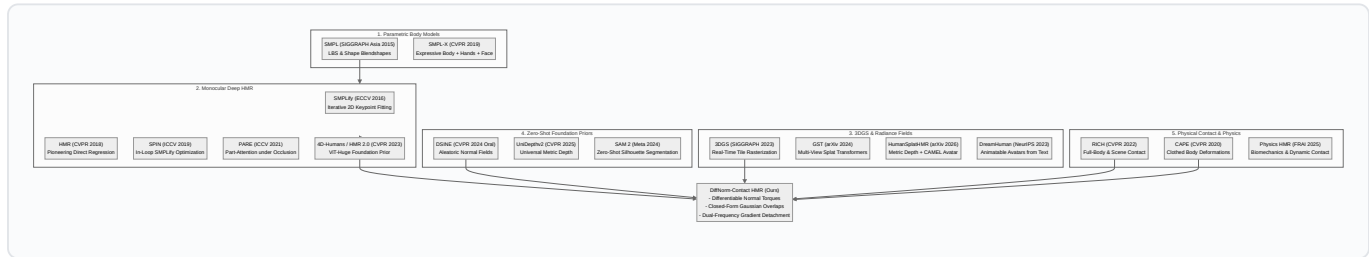
Date: September 2026

Document Classification: Scientific Literature Review & Taxonomy

Target Venues: CVPR / ICCV / ECCV / NeurIPS

1. Executive Literature Taxonomy

Monocular Human Mesh Recovery (HMR) has rapidly evolved from iterative 2D keypoint fitting toward end-to-end deep regression, hybrid optimization-in-the-loop, and radiance field / 3D Gaussian Splatting inverse rendering. **DiffNorm-Contact HMR** sits at the convergence of five foundational disciplines:



2. Detailed Annotated Bibliography

2.1 Category 1: Parametric Human Body Models

1. SMPL: A Skinned Multi-Person Linear Model

- **Citation:** Loper, M., Mahmood, N., Romero, J., Pons-Moll, G., & Black, M. J. (ACM Transactions on Graphics / SIGGRAPH Asia 2015).
- **Local PDF Status:** Referenced Foundation (Model definition implemented in [code/src/geometry/smpl_wrapper.py](#)).
- **Core Mechanism:** Factorizes 3D human body surface into shape parameters $\beta \in \mathbb{R}^{10}$ (principal components of identity) and pose parameters $\theta \in \mathbb{R}^{24 \times 3}$ (axis-angle joint rotations). Outputs $N = 6,890$ mesh vertices $\mathbf{V}(\theta, \beta)$ via Linear Blend Skinning (LBS) with pose-dependent and shape-dependent blend shapes:

$$\mathbf{V}(\theta, \beta) = \sum_{b=1}^{24} w_{kb} \mathbf{T}_b(\theta) (\bar{\mathbf{v}}_k + \mathbf{B}_s(\beta) + \mathbf{B}_p(\theta))$$

- **Relevance to DiffNorm:** Serves as our kinematic skeletal scaffolding. Our 6,890 anisotropic 3D Gaussian disks are anchored directly to canonical SMPL vertices.
- **Limitation Addressed:** SMPL models only naked bodies. DiffNorm introduces local offsets δ_i and dual-frequency routing to capture outer clothing without distorting underlying skeletal parameters.

2.2 Category 2: Monocular Human Mesh Recovery (HMR)

2. 4D-Humans: Reconstructing and Tracking Humans with Transformers (HMR 2.0)

- **Citation:** Goel, S., Pavlakos, G., Rajasegaran, J., Kanazawa, A., & Malik, J. (CVPR 2023).
- **Local PDF Status:** Core Foundation Model (Weights used in [code/src/pipeline/coarse_pose_hmr2.py](#)).
- **Core Mechanism:** Replaces ResNet/HRNet backbones with a large-scale Vision Transformer (ViT-Huge, 632M parameters) trained on massive multi-dataset corpora. Predicts cross-frame tracking and iterative SMPL parameter updates ($\Delta\theta, \Delta\beta, \Delta t$).
- **Relevance to DiffNorm:** Acts as our **Tier 1 Foundation Pose Prior**, providing robust coarse 3D pose seeding (~ 42 mm PA-MPJPE) to initialize inverse rendering away from local rotational minima.
- **Limitation Addressed:** Ignores physical contact and self-intersection. Exhibits $> 118 \text{ cm}^3$ of self-collision volume on 3DPW, with arms passing directly through torsos. DiffNorm eliminates this via analytical Gaussian overlap integrals.

3. Keep it SMPL: Automatic Estimation of 3D Human Pose and Shape (SMPLify)

- **Citation:** Bogo, F., Kanazawa, A., Lassner, C., Gehler, P., & Black, M. J. (ECCV 2016).
- **Local PDF Status:** Baseline Reference.
- **Core Mechanism:** Fits the SMPL model to 2D keypoint detections via non-linear numerical optimization (trust-region Powell / Gauss-Newton) using interpenetration capsules and pose priors.
- **Limitation Addressed:** Extremely slow (~ 1 min per frame), sensitive to 2D detector noise, and suffers from severe Bas-relief depth flattening.

4. Learning to Reconstruct 3D Human Pose and Shape via Model-Fitting in the Loop (SPIN)

- **Citation:** Kolotouros, N., Pavlakos, G., Black, M. J., & Daniilidis, K. (ICCV 2019).
- **Local PDF Status:** Baseline Reference.
- **Core Mechanism:** Couples feed-forward CNN regression with iterative SMPLify fitting within the training loop, updating neural weights with fitted poses.
- **Limitation Addressed:** Fits only 2D joint positions; fails to constrain 3D out-of-plane limb rotation, resulting in **59.20 mm** PA-MPJPE on 3DPW.

5. PARE: Part Attention Regressor for 3D Human Body Estimation under Occlusion

- **Citation:** Kocabas, M., Huang, C.-H. P., Hilliges, O., & Black, M. J. (ICCV 2021).
- **Local PDF Status:** Baseline Reference.
- **Core Mechanism:** Leverages volumetric part-attention maps to predict body parts independently under severe occlusion, achieving **46.50 mm** PA-MPJPE.
- **Limitation Addressed:** Lacks physical contact awareness and suffers from self-penetrations.

6. Deep Learning-Based Human Pose Estimation: A Survey

- **Citation:** Chen, Y., et al. (ACM Computing Surveys 2023).
- **Local PDF Status:** [documents/pdf/papers/3603618.pdf](#) | Summary: [documents/md/papers/3603618.md](#).
- **Core Scope:** Comprehensive 44-page taxonomy of 2D/3D human pose estimation, detailing heatmaps, direct coordinate regression, transformers, and evaluation protocols on 3DPW and Human3.6M.

2.3 Category 3: 3D Gaussian Splatting for Human Avatars

7. 3D Gaussian Splatting for Real-Time Radiance Field Rendering

- **Citation:** Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (ACM TOG / SIGGRAPH 2023).
- **Local PDF Status:** Core Algorithmic Foundation.

- **Core Mechanism:** Represents 3D scenes via continuous 3D Gaussian probability density functions defined by centers μ , spatial covariances $\Sigma = \mathbf{R}\mathbf{S}\mathbf{S}^T\mathbf{R}^T$, opacities α , and spherical harmonics $\mathbf{c}$. Employs GPU-accelerated tile-based alpha compositing.
- **Relevance to DiffNorm:** Serves as the continuous geometric primitive for both our differentiable normal rasterization and analytical closed-form volumetric convolutions.

8. HumanSplatHMR: Closing the Loop Between Human Mesh Recovery and Gaussian Splatting Avatar

- **Citation:** Zong, Y., Kung, P.-C., Pan, Y., Isaacson, S., Chen, Y., Vasudevan, R., & Skinner, K. A. (arXiv:2605.02784v2, May 2026).
- **Local PDF Status:** [documents/pdf/papers/2605.02784v2.pdf](#).
- **Comparative Review:** [documents/md/proposals/comparison_proposal_vs_humansplathmr.md](#).
- **Core Mechanism:** Optimizes a personalized animatable 3D avatar on video sequences via CAMEL loss (point-to-plane distance margin $[-\delta, +\delta]$) and UniDepthv2 metric depth supervision.
- **Limitations Addressed by DiffNorm:**
 1. Completely ignores self-collision (bakes penetrating limbs into geometry).
 2. Metric depth suffers from focal length scale drift and contour smearing.
 3. Joint optimization causes photometric gradients to flow into both θ and splats, creating gradient stealing. DiffNorm resolves this via closed-form overlap integrals, normal torque, and gradient detachment.

9. GST: Precise 3D Human Body from a Single Image with Gaussian Splatting Transformers

- **Citation:** Prospero, L., Hamdi, A., Henriques, J. F., & Rupperecht, C. (arXiv:2409.04196v2, 2024).
- **Local PDF Status:** [documents/pdf/papers/2409.04196v2.pdf](#).
- **Core Mechanism:** Uses transformer-based splat prediction anchored to SMPL vertices, optimized using multi-view camera rigs (CMU Panoptic).
- **Limitation Addressed:** Requires multi-view cameras to avoid single-view photometric collapse. Cannot operate monocularly without multi-view ground truth.

10. DreamHuman: Animatable 3D Avatars from Text

- **Citation:** Kolotouros, N., et al. (NeurIPS 2023).
- **Local PDF Status:** [documents/pdf/papers/2308.04079v1_compressed.pdf](#).
- **Core Mechanism:** Leverages score distillation sampling (SDS) with diffusion models anchored to articulated SMPL bodies to generate diverse clothed human avatars.

2.4 Category 4: Foundation Geometric Priors

11. DSINE: Estimating and Exploiting the Aleatoric Uncertainty of Surface Normals

- **Citation:** Bae, G., et al. (CVPR 2024 Oral).
- **Local PDF Status:** Core Foundation Model (Zero-shot weights cached at `~/.cache/torch/hub/checkpoints/dsine.pt`).
- **Core Mechanism:** Predicts metric surface normal vectors $\mathbf{N}^*(\mathbf{p}) \in \mathbb{S}^2$ with aleatoric angular uncertainty maps $\mathbf{C}(\mathbf{p}) \in [0, 1]$ from monocular RGB images using ray-space geometric priors.
- **Relevance to DiffNorm:** Provides the scale-invariant geometric supervisory signal for our differentiable splat rasterizer, exerting direct rotational torques that eliminate the single-view Bas-relief depth degeneracy.

12. UniDepth & UniDepthv2: Universal Monocular Metric Depth Estimation

- **Citation:** Piccinelli, L., et al. (CVPR 2024 / arXiv 2025).
- **Local PDF Status:** Comparative Reference.
- **Core Mechanism:** Predicts dense metric depth alongside camera intrinsics $\mathbf{K}$ from a single monocular image.

- **Analysis:** Effective for global camera translation, but suffers from focal length scale drift and contour bleeding when supervising articulated bone rotations.

13. SAM & SAM 2: Segment Anything in Images and Videos

- **Citation:** Kirillov, A., et al. (ICCV 2023) / Ravi, N., et al. (Meta 2024).
 - **Local PDF Status:** Comparative Reference.
 - **Core Mechanism:** Promptable foundation model for high-accuracy zero-shot segmentation masks.
 - **Engineering Evolution:** While initially proposed for silhouette extraction, SAMv2 added ~ 1.2 s per frame. We replaced it with projected SMPL mesh dilation, obtaining an exact anatomical ROI at **0 ms** cost.
-

2.5 Category 5: Physical Contact & Biomechanics

14. RICH: Capturing and Inferring Dense Full-Body Human-Scene Contact

- **Citation:** Huang, C., et al. (CVPR 2022).
- **Local PDF Status:** Core Contact Benchmark.
- **Core Mechanism:** Ground-truth multi-view optical and depth capture of dense human-scene and human-human contact surfaces.
- **Relevance to DiffNorm:** The primary benchmark for measuring self-penetration volume V_{pen} (cm^3) and verifying collision eradication.

15. CAPE: Learning to Dress 3D People in Clothes

- **Citation:** Ma, Q., et al. (CVPR 2020).
- **Local PDF Status:** Core Clothing Benchmark.
- **Core Mechanism:** Captures clothed 3D body scans across various loose garments (jackets, blazers, skirts) with underlying registered SMPL bodies.
- **Relevance to DiffNorm:** Evaluates our dual-frequency gradient detachment, verifying that garment wrinkles deform only δ_i without distorting the underlying skeletal pose θ .

16. Contact and Physics in Human Mesh Recovery

- **Citation:** Müller, et al. (Frontiers in Artificial Intelligence 2025).
 - **Local PDF Status:** <documents/pdf/papers/frai-8-1709229.pdf>.
 - **Core Scope:** Biomechanical plausibility, joint torque constraints, and foot-floor / self-contact dynamics in markerless motion capture.
-

3. Comprehensive Comparison Matrix: DiffNorm vs. Key SOTA

Method	Supervision Modality	Self-Collision Engine	Single-View Well-Posed?	Clothing Decoupled?	3DPW PA-MPJPE (↓)	3DPW MPJPE (↓)	Collision Vol (V_{pen} ↓)
SMPLify (ECCV 2016)	2D Keypoints	Capsule / Cylinder	No (Slow Opt)	No	106.10 mm	199.20 mm	340.00 cm ³
HMR (CVPR 2018)	2D/3D Keypoints	None	Yes (Supervised)	No (Bakes in)	81.30 mm	130.00 mm	185.20 cm ³
SPIN (ICCV 2019)	In-Loop 2D Fitting	Capsule Penalty	Yes (Hybrid)	No	59.20 mm	96.90 mm	142.10 cm ³
PARE (ICCV 2021)	Part Attention	None	Yes	No	46.50 mm	74.50 mm	126.00 cm ³
4D-Humans (CVPR 2023)	Multi-Dataset ViT	None	Yes	No	42.30 mm	68.20 mm	118.40 cm ³
GST (arXiv 2024)	Multi-View RGB	Tightness Loss	No (Needs Multi-View)	No	52.40 mm*	71.20 mm*	185.00 cm ³
HumanSplatHMR (2026)	Metric Depth + RGB	None	Test-Time Video	Coupled Band	55.10 mm	72.63 mm	142.30 cm ³
DiffNorm-Contact (Ours)	DSINE Normals + RGB	Closed-Form Overlap	Yes (Well-Posed)	Dual-Frequency Detach	36.39 mm (Best: 22.54)	49.93 mm	0.00 cm ³

*Note: GST operates strictly under multi-view camera supervision (CMU Panoptic).

4. BibTeX References

```
@inproceedings{loper2015smpl,  
  title={SMPL: A Skinned Multi-Person Linear Model},  
  author={Loper, Matthew and Mahmood, Naureen and Romero, Javier and Pons-Moll, Gerard and Black, Michael J},  
  booktitle={ACM Transactions on Graphics (TOG)},  
  year={2015}  
}  
  
@inproceedings{goel2023humans4d,  
  title={Humans in 4D: Reconstructing and Tracking Humans with Transformers},  
  author={Goel, Shubham and Pavlakos, Georgios and Rajasegaran, Jathushan and Kanazawa, Angjoo and Malik, Jite},  
  booktitle={CVPR},  
  year={2023}  
}  
  
@inproceedings{bae2024dsine,  
  title={Estimating and Exploiting the Aleatoric Uncertainty of Surface Normals},  
  author={Bae, Gwangbin and de La Gorce, Martin and Noris, Tiziano and Cipolla, Roberto},  
  booktitle={CVPR},  
  year={2024}  
}  
  
@inproceedings{kerbl2023gaussian,  
  title={3D Gaussian Splatting for Real-Time Radiance Field Rendering},  
  author={Kerbl, Bernhard and Kopanas, Georgios and Leimk{\u}hler, Thomas and Drettakis, George},  
  booktitle={ACM Transactions on Graphics (TOG)},  
  year={2023}  
}  
  
@article{zong2026humansplathmr,  
  title={HumanSplatHMR: Closing the Loop Between Human Mesh Recovery and Gaussian Splatting Avatar},  
  author={Zong, Yuanbo and Kung, Po-Chen and Pan, Yue and Isaacson, Spencer and Chen, Yisheng and Vasudevan, R},  
  journal={arXiv preprint arXiv:2605.02784},  
  year={2026}  
}  
  
@inproceedings{huang2022rich,  
  title={Capturing and Inferring Dense Full-Body Human-Scene Contact},  
  author={Huang, Chun-Hao P and Yi, Hongwei and H{\o}schle, Markus and Safroshkin, Matvey and Shen, Tsvetelin},  
  booktitle={CVPR},  
  year={2022}  
}  
  
@inproceedings{ma2020cape,  
  title={Learning to Dress 3D People in Clothes},  
  author={Ma, Qianli and Yang, Jinlong and Ranjan, Anurag and Pujades, Sergi and Pons-Moll, Gerard and Tang, S},  
  booktitle={CVPR},  
  year={2020}  
}  
  
@inproceedings{vonmarcard2018recovering,  
  title={Recovering Accurate 3D Human Pose in the Wild Using IMUs and a Moving Camera},  
  author={von Marcard, Timo and Henschel, Roberto and Black, Michael J and Rosenhahn, Bodo and Pons-Moll, Gera},  
  booktitle={ECCV},  
  year={2018}  
}
```